CHAPTER 1

The Property and Capital Markets

What is real estate? How big is the real estate sector? How does the market for the use of real estate differ from the market for real estate assets? In this chapter, our objectives are (1) to define *real estate,* (2) to describe how the real estate sector operates within the national economy, and (3) to distinguish between the market for real estate use (where space is rented or purchased for occupancy) and the market for real estate assets (where buildings are bought and sold as investments).

The most common definition of *real estate* is the national stock of buildings, the land on which they are built, and all vacant land. These buildings are used either by firms, government, nonprofit organizations, and so on, as workplaces, or by households as places of residence. When defined this way, the value of all real estate makes up the largest single component of national wealth.

The yearly value of new buildings put in place represents an annual investment in the nation's stock of capital. The dollar value of these new buildings also has been the largest single category of national investment in recent years. Investment in new buildings accounts for roughly 7 percent of gross domestic product (GDP). Of this 7 percent, about 60 percent is payments to the construction sector (for labor, construction equipment, etc.), with the remaining 40 percent going to the producers of building materials. It is important to note that land is not counted as part of investment or GDP since it is not a produced commodity. On the other hand, land is a national asset, and so the value of land beneath buildings should be counted as stock or wealth.

The distinction between real estate as space and real estate as an asset is most clear when buildings are not occupied by their owners. The needs of tenants and the type and quality of buildings available determine the rent for space in the market for property use. At the same time, buildings may be bought, sold, or exchanged between investors. These transactions occur in the capital market and determine the *asset* price of space. In this chapter, we present a simple analytic framework which illustrates the connections between the market for real estate space (the property market) and the market for real estate assets (the asset or capital market). As we will show later in this chapter, this same approach can be used for owner-occupied real estate where the user of space is also the investor in the asset.

Our view of the real estate market as actually two markets helps to clarify how different forces influence this important sector. If, for example, there is a sudden demand by foreign investors to purchase U.S. office buildings, the impact on rents is very different than if firms suddenly decide that they wish to rent more office space for their use. A reduction in long-term mortgage rates has just the opposite effect on house prices from that caused by a reduction in short-term rates for construction financing. Distinguishing between the property and capital markets provides a clearer understanding of how such forces operate in the real estate sector as a whole.

In addition, there is a methodological objective in this first chapter—to reacquaint the reader with simple supply and demand analysis. By considering how the markets for both real estate property and assets operate within a global economy, we recall the distinction between endogenous economic variables and exogenous economic forces.[1] Within the markets for real estate, economic variables like prices, sales, and output are all determined endogenously. The outcome of the market, however, may be strongly influenced by exogenous forces such as world trade, interest rates, or even climate. Studying how changes in exogenous forces affect endogenous variables is one of the most important and fundamental pursuits of economics. How this methodology is applied to the real estate markets is a major focus of this book.

THE SIZE AND CHARACTER OF U.S. REAL ESTATE

Private real estate is composed of all types of buildings as well as the land they sit on. Houses, office buildings, warehouses, and shopping centers all are clear examples of real estate. Other, less obvious examples include privately owned ice-skating rinks and aircraft hangars. These are the buildings that various service firms (e.g., recreation, lodging, and travel) need for their operations. Finally, petroleum refineries, steel mills, and utility power plants are partially real estate—some portion of these structures is considered industrial "plant," while the remainder is "equipment." Public real estate is composed of all government-owned office buildings, schools, firehouses, military barracks, and the like.

[1]By *exogenous forces* we mean factors that influence real estate market outcomes but are not influenced by the real estate market. For example, interest rates have profound impacts on real estate market outcomes, but the real estate market has very little if any impact on interest rates. *Endogenous variables* are measures of real estate market outcomes. Prices and rents are endogenous, which means that they are determined by the real estate market.

Like any economic variable, real estate can be measured both as a flow and as a stock. The *flow* of real estate is the value of new buildings put in place each year, less losses from the stock through depreciation or demolition. New completions represent an important component of national investment, which is also a flow variable. On the other hand, the total value of all existing buildings and the value of all land are *stock* variables, and are part of national wealth—also a stock variable. Since land is nonreproducible, it is always a stock variable and never a flow variable.

Table 1.1 breaks down the value of new construction put in place in 1990. Virtually all private construction was in the form of buildings and represented $301 billion (5.5

TABLE 1.1 Value of New Construction Put in Place, 1990

	$ (in billions)	% of GDP
Private Construction	338	6.1
Buildings	301	5.5
Residential Buildings	183	3.3
Nonresidential Buildings	118	2.1
Industrial	24	0.4
Office	29	0.5
Hotels/Motels	10	0.2
Other Commercial	34	0.6
All Other Nonresidential	21	0.4
Nonbuilding Construction	37	0.7
Public Utilities	31	0.6
All Other	6	0.1
Public Construction	109	2.0
Buildings	46	0.8
Housing and Development	4	0.1
Industrial	1	0.0
Other	41	0.7
Nonbuilding Construction	63	1.1
Infrastructure	55	1.0
All Other	8	0.1
Total New Construction	446	8.1
Total GDP:	5,514	100.0

Source: Current Construction Reports, Series C30, U.S. Bureau of the Census; gross domestic product, *Economic Report of the President 1992.*

percent of GDP). Residential buildings accounted for about 61 percent of the dollar value of private buildings constructed, with office, industrial and other commercial structures representing the remainder. In the public sector, buildings represented only about 42 percent of the $109 billion in public construction. The largest component of public construction was investment in infrastructure (roads, bridges, airports, etc.).

Total gross private domestic investment, which measures purchases of fixed capital goods including structures and equipment, was $803 billion in the U.S. in 1990. (Board of Governors 1991). In these figures, structures include purchases of new residential and nonresidential structures, as well as net purchases of existing structures. The structure component accounted for just over half of the nation's gross investment ($415 billion), while the remaining $388 billion was investment in machinery or equipment.

Over the years, government statistics have tracked the flow of gross investment into real estate (i.e., new building construction) with a high degree of accuracy. Valuing the total real estate stock at any point in time, however, is far more difficult. The Bureau of Economic Analysis (BEA) provides estimates of "fixed reproducible tangible wealth," which attempt to value the stock of both residential and nonresidential structures. Estimates of gross stock are calculated by summing investment flows and deducting the cumulated investment in structures that have been discarded. The investment flow data are from the national income and product accounts. Net capital stock is calculated by using a depreciation formula to write off the value over the life of the stock. The Federal Reserve uses the BEA estimates of net stock and adds their estimates of the value of land to provide balance sheet estimates of the value of U.S. real estate (Miles 1990).

An important problem with this approach to value is that the cumulated investment flow less demolition and depreciation may bear little resemblance to the actual market value of real estate. While the investment flow data may be a fairly accurate measure of the value of additions to the stock in the current year, changes in the value of those additions to the stock over time are a function of market forces that are unlikely to be captured even by the most sophisticated estimates of depreciation. In addition, it is difficult to estimate the value of the land on which those structures are built because most observed transactions provide a single purchase price for the existing building and the land, with no breakdown by the land and structure components.

A 1991 study made a gallant attempt at estimating the value of all U.S. real estate according to type of real estate (e.g., residential, office, and retail) and type of owner (e.g., individuals, corporations) by piecing together data from a variety of sources (IREM 1991). The study employed standard government statistics, such as the American Housing Survey (AHS) for residential units, and trade association data such as those collected by the International Council of Shopping Centers. In addition, it used state and county property tax records as a basis for regression models that estimate real estate value. Finally, the study relied on interviews with industry experts to augment their data.

In this study, total real estate in the U.S. for 1990 was estimated to be worth $8.8 trillion. As shown in Table 1.2, almost 70 percent of all U.S. real estate was residential, and almost 90 percent of the value of residential real estate was in the nation's stock of single-family homes. The 30 percent of U.S. real estate that was nonresidential was dominated by office and retail space—at least in dollar value. While estimates of total national

TABLE 1.2 Value of U.S. Real Estate, 1990

	$ (in billions)	% of Total
Residential	6,122	69.8
Single-Family Homes	5,419	61.7
Multifamily	552	6.3
Condominiums/Coops	96	1.1
Mobile Homes	55	0.6
Nonresidential	2,655	30.2
Retail	1,115	12.7
Office	1,009	11.5
Manufacturing	308	3.5
Warehouse	223	2.5
Total U.S. Real Estate	8,777	100.0

Source: Managing the Future: Real Estate in the 1990s (Chicago: IREM Foundation and Arthur Anderson, 1991), pp. 28, 31.

wealth often vary, the Federal Reserve estimates that national net worth was at $15.6 trillion in 1990. Thus, the figures in Table 1.2 suggest that real estate, valued at $8.8 trillion, constituted roughly 56 percent of the nation's wealth in 1990.[2]

Who owns American real estate? The legal ownership status of this wealth is shown in Table 1.3. In 1990, 83 percent of residential real estate was owned by individuals. It is important to remember that this figure covers not only individual ownership of personal residences but also sole-proprietor ownership of apartment buildings. Similarly, the 62 percent of nonresidential real estate that was owned by corporations consists of investment property as well as buildings occupied by their corporate owners. Partnerships own almost equal shares of residential and nonresidential property. Finally, the ownership of U.S. real estate by foreign entities is very small, despite considerable concern about this during the last few years.

In summary, U.S. real estate is the largest single component of national wealth and the largest component of annual net private investment. This huge base of assets, however, has been accumulated by devoting only about 5 to 7 percent of each year's GDP to the construction and renovation of that base. It is, of course, the durability of real estate that allows us to devote such a small fraction of GDP to the accumulation and maintenance of such a large share of our assets.

[2]The national net worth estimate is from *Balance Sheets for the U.S. Economy 1945–1990* (Washington, D.C.: Board of Governors of the Federal Reserve System, 1991). It should be noted that the Federal Reserve estimates the value of real estate at $10.7 trillion. Miles (1991:74) argues that the BEA/Federal Reserve estimates of the value of nonresidential real estate may be high because the data used for the stock estimates include special fixtures in manufacturing plants that are certainly part of the capital stock, but should not be included when measuring the value of real estate.

TABLE 1.3 Who Owns U.S. Real Estate, 1990

	All Real Estate		Residential Only		Nonresidential Only	
	$ (in billions)	%	$ (in billions)	%	$ (in billions)	%
Individuals	5,088	58.0	5,071	82.8	17	0.6
Corporations	1,699	19.4	66	1.1	1,633	61.5
Partnerships	1,011	11.5	673	11.0	338	12.7
Nonprofits	411	4.7	104	1.7	307	11.6
Government	234	2.6	173	2.8	61	2.3
Institutional Investors	128	1.5	14	0.2	114	4.3
Financial Institutions	114	1.3	13	0.2	101	3.8
Other (Includes Foreign)	92	1.0	8	0.1	84	3.2
Total:	8,777	100.0	6,122	100.0	2,655	100.0
% of All Real Estate:		100.0		69.8		30.2

Source: Managing the Future: Real Estate in the 1990s (Chicago: IREM Foundation and Arthur Anderson, 1991), pp. 29–33.

THE MARKETS FOR REAL ESTATE ASSETS AND REAL ESTATE USE

Since real estate is a durable capital good, its production and price are determined in an asset, or capital, market. In this market, the demand to own real estate assets must equal their supply. Thus, the price of houses in the U.S. largely depends on how many households wish to *own* units and how many units are available for ownership. Likewise, the value or price of shopping center space depends on how many investors wish to own such space and how many centers there are available to invest in. In both cases, all else being equal, an increase in the demand to own these assets will raise prices, while a greater supply of space will depress prices.

The supply of new real estate assets comes from the construction sector and depends on the price of those assets relative to the cost of replacing or constructing them. In the long run, the asset market should equate market prices with replacement costs that include the cost of land. In the short run, however, the two may diverge significantly because of the lags and delays that are inherent in the construction process. For example, if demand for the ownership of space suddenly rises, then, with a fixed supply of assets, prices will rise as well. With prices now above construction and land costs, new development takes place. As this space arrives on the market, demand is satisfied and prices begin to fall back towards the cost of replacement. A question to be addressed in future chapters is whether the cost of asset replacement is constant, varies with the level of development, or depends on the total stock of assets.

What would cause the demand for owning real estate assets to suddenly increase? More generally, are there other determinants of asset demand besides simply the price of these assets? The answer is yes, and the most important of these determinants is the rental income that real estate assets earn. To understand rent, it is necessary to consider the market for the *use* of real estate.

In the market for real estate use or space (referred to here as the property market), demand comes from the *occupiers* of space, whether they be tenants or owners, firms or households. For firms, space is one of many factors of production, and, like any other factor, its use will depend on firm output levels and the relative cost of space. Households likewise divide their income into the consumption of many commodities, only one of which is space. The household demand for space depends on income and the cost of occupying that space relative to the cost of other commodities, such as food, clothing, or entertainment. For firms or households, the cost of occupying space is the annual outlay necessary to obtain the use of real estate—its *rent.* For tenants, rent is simply specified in a lease agreement. For owners, rent is defined as the annualized cost associated with the ownership of property.

Rent is determined in the property market for space use, not in the asset market for ownership. In the property market, the supply of space is given (from the asset market). The demand for space depends on rent and other exogenous economic factors such as firm production levels, income levels, or the number of households. The task of the property market is to determine a rent level at which the demand for space use equals the supply of space. All else being equal, when the number of households increases or firms expand production, the demand for space *use* rises. With fixed supply, rents rise as well.

The link between the asset market and the property market occurs at two junctions. First, the rent levels determined in the property market are central in determining the demand for real estate assets. After all, in acquiring an asset, investors are really purchasing a current or future income stream. Thus, changes in rent occurring in the property market immediately affect the demand for ownership in the asset market. The second link between the two markets occurs through the construction or development sector. If construction increases and the supply of assets grows, not only are prices driven down in the asset market, but rents decline in the property market as well. These connections between the two markets are illustrated in the four-quadrant diagram in Figure 1.1.

In explaining Figure 1.1, it is useful to refer to quadrants by their compass designation. The two right-hand quadrants (northeast and southeast) represent the property market for the use of space, while the two left-hand quadrants (northwest and southwest) deal with the asset market for the ownership of real estate. Let's begin with the northeast quadrant, where rents are determined in the short run.

The northeast quadrant has two axes: rent (per unit of space) and the stock of space (also measured in units of space, such as square feet). The curve represents how the demand for space depends on rents, given the state of the economy. Movement along that curve depicts how much space would be demanded given a particular rent level on the vertical axis. If households or firms tend to demand the same amount of space regardless of rent levels (inelastic demand), then the curve is nearly vertical. If space usage is quite sensitive to rents (elastic demand), then the curve is more horizontal. If the economy changes, then the entire curve shifts. An upward shift occurs with an increase in firms or households (economic growth) and signifies that more space is demanded for the same rent. Economic decline causes a downward shift in the line, with the reverse implications.

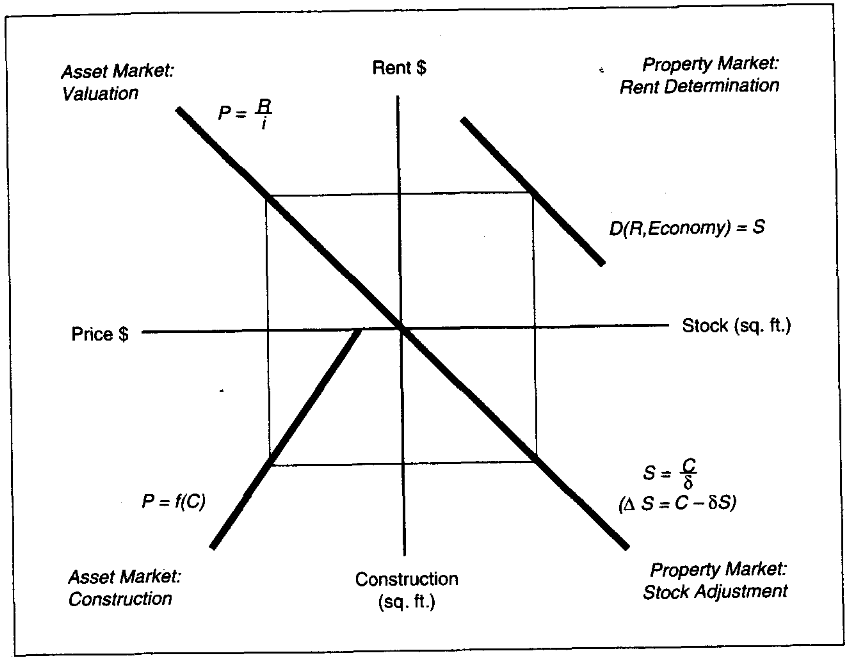

FIGURE 1.1 Real estate: the property and asset markets.

In equilibrium the demand for space, D, is equal to the stock of space, S. Thus, rent, R, must be determined so that demand is exactly equal to stock. Demand is a function of rent and conditions in the economy:

$$D(R, \text{Economy}) = S \tag{1.1}$$

Recall that in the property market the supply of stock is given from the asset market. In Figure 1.1, then, rent is determined by taking a level of stock of space on the horizontal axis, drawing a line up to the demand curve, and then over to the vertical axis. With this rent for the use of space, we then move to the asset market in the northwest quadrant.[3]

The northwest quadrant represents the first part of the asset market and has two axes: rent and price (per unit of space). The ray emanating from the origin represents the

[3]If we take the office market as an example, it would be reasonable to assume that the demand for office space use (in square feet), D, is equal to: $E(400 - 10R)$, where E is the number of office workers in the economy (in millions), and R is the annual rent per square foot. In the northeast quadrant, we equate this demand to the existing stock, S, solving for rent. Equation (1.1) can be rewritten as:

$$R = 40 - \frac{S}{10E}$$

capitalization rate for real estate assets: the ratio of rent-to-price. This is the current yield that investors demand in order to hold real estate assets. Generally, four considerations make up this capitalization rate: the long-term interest rate in the economy, the expected growth in rents, the risks associated with that rental income stream, and the treatment of real estate in the U.S. federal tax code. A higher capitalization rate is represented by a clockwise rotation in the ray, while a decline in the cap rate is represented by a counter-clockwise rotation. In this quadrant, the capitalization rate is taken as exogenous, based on interest rates and returns in the broader capital market for all assets (stocks, bonds, short-term deposits). Thus, the purpose of the northwest quadrant is to take the rent level, R, from the northeast quadrant and determine a price for real estate assets, P, using a capitalization rate, i:

$$P = \frac{R}{i} \tag{1.2}$$

This is done by moving from the rent level on the vertical axis in the northeast quadrant over to the ray in the northwest quadrant, and then down to the horizontal axis where asset price is given.

The next (southwest) quadrant is that portion of the asset market in which the creation of new assets is determined. Here, the curve $f(C)$ represents the replacement cost of real estate. In this version of the diagram, the cost of replacement through new construction is assumed to increase with greater building activity (C), and so the curve moves in a southwesterly direction. It intersects the price axis at that minimum dollar value (per unit of space) required to get some level of new development underway. If this construction can be supplied at any level with almost the same costs, then the ray will be close to vertical. Construction bottlenecks, scarce land, and other impediments to development lead to inelastic supply and a ray that is more horizontal. Given the price of real estate assets from the northwest quadrant, a line down to the replacement cost curve and then over to the vertical axis determines the level of new construction where replacement costs equal asset prices.[4] Lower levels of construction would lead to excess profits, whereas higher levels would be unprofitable. Hence, new construction occurs at that level, C, at which asset price, P, is equal to replacement costs, $f(C)$:

$$P = f(C) \tag{1.3}$$

In the southeast quadrant, the annual flow of new construction, C, is converted into a long-run stock of real estate space. The change in the stock, ΔS, in a given period is equal to new construction minus losses from the stock measured by the depreciation (removal) rate, δ:

$$\Delta S = C - \delta S \tag{1.4}$$

[4]For the office market example, a reasonable cost function (dollar per square foot) for the development of new office space would be: $200 + 5C$. The annual level of new construction, C, is in millions of square feet. If these costs are equated to the asset price (per square foot) of office space, Equation (1.3) can be rewritten to solve for the level of construction where costs equal asset prices:

$$C = \frac{P - 200}{5}$$

The ray emanating from the origin represents that level of stock (on the horizontal axis) that requires an annual level of construction for replacement just equal to that value on the vertical axis. At that level of stock and corresponding level of construction, the stock of space will be constant over time, since depreciation will equal new completions. Hence, ΔS is equal to 0 and $S = C/\delta$.[5] Future chapters will discuss this relationship in more detail; for now, it is important only to remember that the southeast quadrant assumes a certain level of construction and determines the level of stock that would result if that construction continued forever.

This completes a 360-degree rotation around the four-quadrant diagram. Starting with a level of stock, the property market determines rents, which are then translated into property prices by the asset market. These asset prices, in turn, generate new construction, which, back in the property market, eventually yields a new level of stock. The combined property and asset markets are in equilibrium when the starting and ending levels of stock are the same. If the ending stock differs from the starting stock, then the values of the four variables in the diagram (rent, prices, construction, and stock) are not in complete equilibrium. If the starting value exceeds the finishing value, then rents, prices, and construction must all rise to be in equilibrium. If the initial stock is less than the finishing stock, then rents, prices, and construction must be decreased to be in equilibrium. This journey around the four-quadrant diagram provides a simple, intuitive illustration of the solution to the simultaneous system of Equations (1.1)–(1.4).[6]

OWNER-OCCUPIED REAL ESTATE

A reasonable question is how all of this works in the case of real estate that is mainly occupied by its owner. In this case, the four quadrants still hold, but asset prices and rents are determined by the same market participants—the owner occupants. Consider for the

[5]The increase each year in the stock of office space is the difference between construction and depreciation. Assuming depreciation, δ, is 1 percent annually, then: $\Delta S = C - 0.01S$. Given a stable level of construction, the stock that will eventually emerge is: $S = C/0.01$. This is the steady-state version of Equation (1.4) and is the ray emanating out of the southeast quadrant.

[6]If the equations in footnotes 3–5 plus Equation (1.2) are successively substituted into each other, the result is:

$$S = \frac{800 - 2\frac{S}{E}}{i} - 4{,}000$$

Solving so that S is only on the left-hand side of the equation, we get the equilibrium stock level:

$$S = E\frac{[800 - 4{,}000i]}{iE + 2}$$

Thus, if there are 10 million office workers in the economy ($E = 10$) and the interest or capitalization rate is 5% ($i = 0.05$), the long-run stock of office space will be 240 square feet per worker, or 2,400 million square feet. This stock will require that 24 million square feet be constructed each year, and this will require that asset prices equal a replacement cost of \$320 per square foot. Rents of \$16 annually will sustain such asset prices with a 5% capitalization rate.

moment the market for owner-occupied housing. The demand for single-family homes depends on the number of households, their incomes, and the annual costs of owning a home. This annual cost is equivalent to rent. A rise in the number of households shifts the demand curve out. With greater demand and a fixed stock of housing units, the annual payment to occupy a house must rise. The northwest quadrant then translates this payment into the actual price that households are willing to pay for the home.

Lower interest rates, for example, imply that with the same annual payment (rent), households can afford to pay a higher purchase (asset) price. Hence, with owner-occupied real estate, decisions by the user-owner determine both rent (the annual payment) and price. These decisions, however, are influenced by the same economic and capital market conditions that influence rental properties. Thus, owner-occupiers have the same investment motives as the owners of rental property. Once the purchase price is determined, then new housing development and eventually a new equilibrium stock of space follow in the other two quadrants (southwest, southeast).

This same analysis can be applied to the case of corporate-owned and -occupied industrial or office space. The demand for this space is determined by the annual cost of owning it (rent) as well as the number and size of firms in the market. In equilibrium, the annual cost of owning (rent) equates demand with the fixed stock of office or industrial space. The cost of capital for the corporation (capitalization rate) converts this annual cost into the asset price that corporations are willing to pay for the space.

REAL ESTATE AND THE NATIONAL ECONOMY

Using Figure 1.1, we can trace the various impacts of the broader economy on the real estate market. The economy can grow or contract. Long-term interest rates or other factors can shift the demand for real estate assets. Changes in short-term credit availability or local regulations can alter the cost of supplying new space. Each has different repercussions, and these are easily determined by examining alternative solutions within the four-quadrant diagram. In each case, we can identify which quadrant is initially affected, trace the impacts through the other quadrants, and arrive at a new long-run equilibrium. This comparison of different long-run solutions (market equilibria) in a model is called "comparative static" analysis.

Economic Growth and the Demand for Real Estate Use

As the economy expands, the curve in the northeast quadrant shifts out to the northeast. This reflects a greater demand to use space at current (or other) rent levels, such as would occur with increases in production, household income, or the number of households. For a given level of real estate space, rents must therefore rise if the demand to use space is to equal available space. These higher rents then lead to greater asset prices in the northwest quadrant, which, in turn generate a higher level of new construction in the southwest quadrant. Eventually, this leads to a greater stock of space (southeast quadrant). As shown

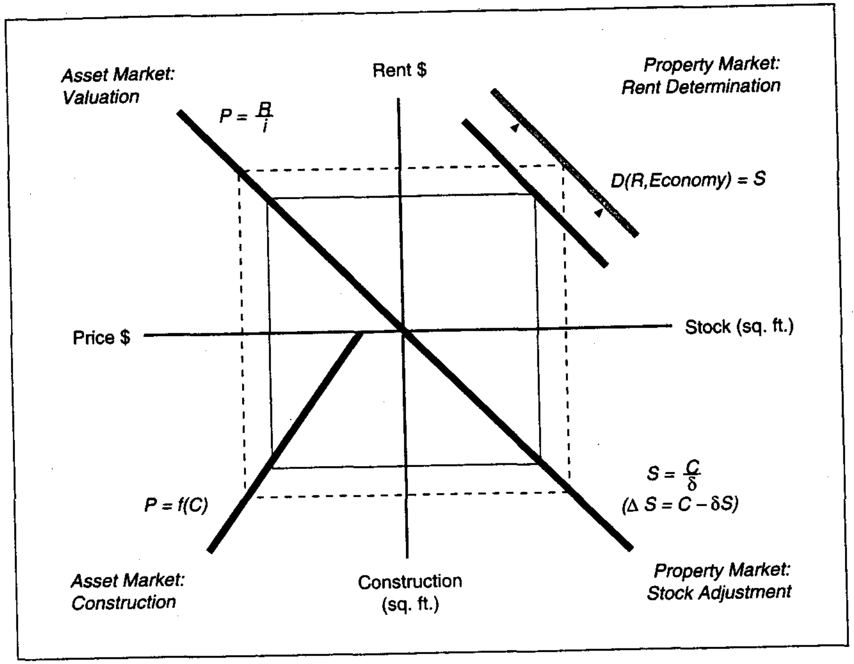

FIGURE 1.2 The property and asset markets: property demand shifts.

in Figure 1.2, the new market equilibrium involves a dashed box that in every direction lies outside of the box that connected the four curves in the original equilibrium.[7]

It should be clear that the equilibrium solution with an expanded economy must generate a box that lies outside of the original. Neither rents, prices, construction, nor stock can be less than in the initial equilibrium. The new solution, however, need not at all involve a proportional expansion of the original box. The shape of the new box depends on the slopes of the various curves. For example, if construction were very elastic with respect to asset prices (a nearly vertical curve in the southwest quadrant), then the new levels of prices and rents would be only slightly greater than before, whereas construction and stock would expand considerably.

[7]We can use growth in the number of office workers as an example of economic expansion. Returning to the equation in footnote 6, if the number of office workers increased from 10 million to 20 million, the long-run stock of space would increase to 4,000 million square feet, or just 200 square feet per worker. Rents would increase to $20 per square foot annually. Such rents would generate asset prices of $400 per square foot, which would just equal the replacement costs necessary to sustain the higher level of annual construction (40 million sq. ft.).

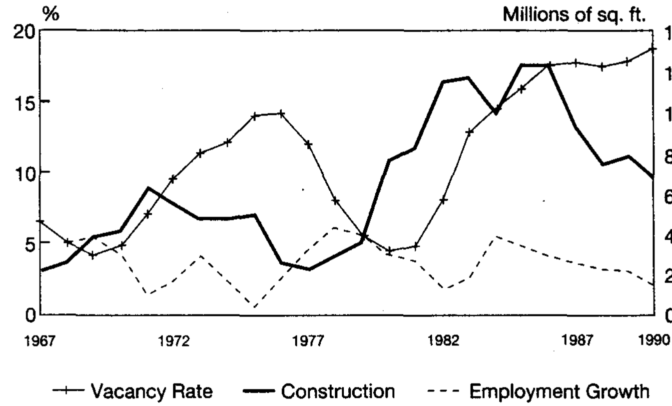

FIGURE 1.3 Office employment growth, vacancy rate, and construction, 1967–1990.

These data are aggregated from 30 metropolitan areas.

Source: Employment, adjusted U.S. government figures courtesy of Regional Financial Associates, Bala-Cynwyd, PA; vacancy and construction, CB Commercial.

Economic growth, then, increases all equilibrium variables in the real estate market(s), while economic contraction leads to decreases in all variables. Figure 1.3 compares the growth of total employment in the U.S. with construction of office space and the overall office vacancy rate. It is clear that the national office market moves with the economy: during recessions, vacancies tend to rise and construction falls, whereas the opposite occurs during recoveries.

Long-Term Interest Rates and the Demand for Real Estate Assets

If the demand to *own* real estate shifts, the impact on the combined markets is quite different than if the demand to *use* real estate changes. A number of factors can cause shifts in the demand to own real estate assets. If interest rates in the rest of the economy rise (fall), then the existing yield from real estate becomes low (high) relative to fixed income securities and investors will wish to shift their funds from (into) the real estate sector. Similarly, if the risk characteristics of real estate are perceived to have worsened (improved), then the existing yield from real estate may also become insufficient (more than necessary) to get investors to purchase real estate assets relative to other assets. Finally, changes in how real estate income is treated in the U.S. federal tax code can also greatly impact the demand to invest in real estate. As will be discussed in Chapter 8, if the depreciation allowance for real estate is increased, the same income stream generates a higher after-tax yield. This will increase the demand to hold real estate assets.

This book assumes that the capital market efficiently adjusts the prices of particular assets—so that each investment earns a common risk-adjusted, after-tax total rate of return. Thus, shifts in asset demand, such as described above, will alter the capitalization rate at which investors are willing to hold real estate. Reductions in long-term interest rates, decreases in the perceived risk of real estate, and generous depreciation or other favorable changes in the tax treatment of real estate all will reduce the yield that investors require from real estate. As shown in Figure 1.4 in the northwest quadrant, this has the effect of a counterclockwise rotation (the ray always goes through the origin) in the capitalization rate ray emanating from the origin, raising asset prices. Higher interest rates, greater perceived risk, and adverse tax changes rotate the ray in a clockwise manner, lowering asset prices.

Given a level of rent from the property market, a reduction in the current yield or capitalization rate for real estate raises asset prices, and in the southwest quadrant, this begins to expand construction. Eventually this increases the stock of space (in the southeast quadrant), which then lowers rents in the property market for space (northeast quadrant). A new equilibrium requires that the initial and finishing rent levels be equal. In

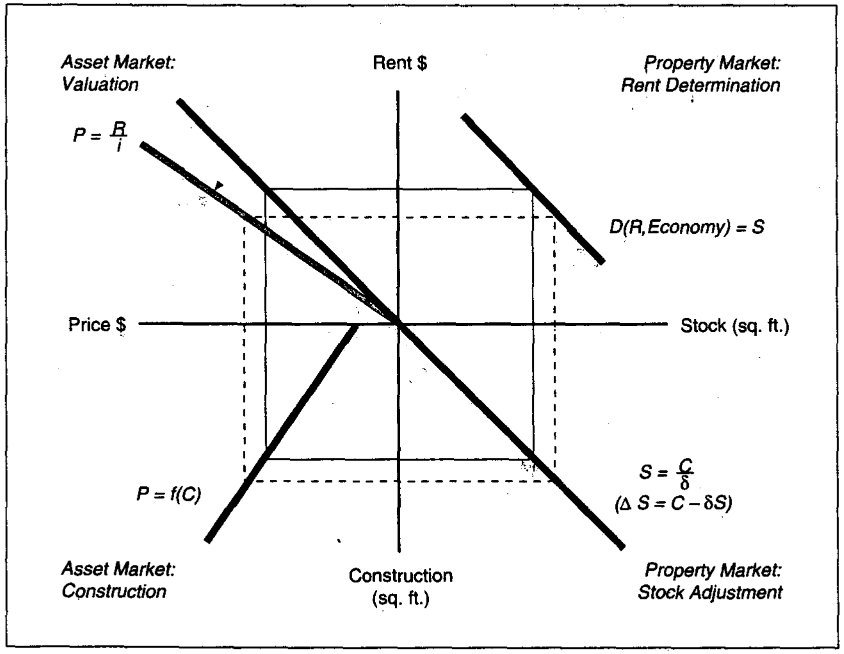

FIGURE 1.4 The property and asset markets: asset demand shifts.

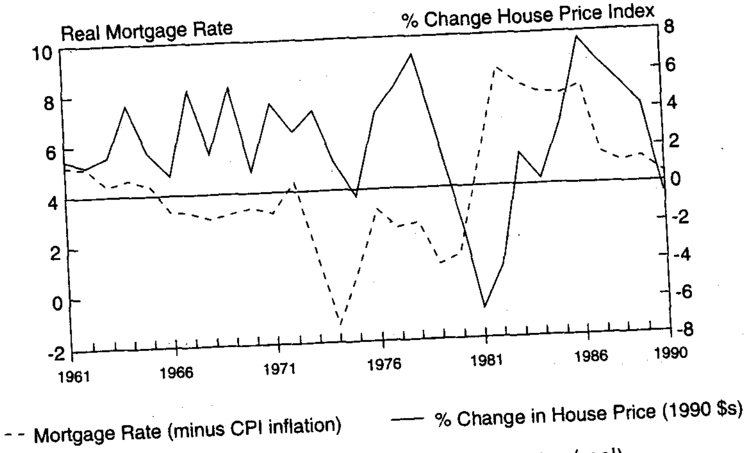

FIGURE 1.5 Change in house price versus mortgage rates (real), 1961–1990.

Source: Price index: 1960–69: Federal Housing Finance Board, *1990 Rates & Terms on Conventional Home Mortgages, Annual Summary.* 1970–90: Federal Home Loan Mortgage Corp., *Quality-Controlled Existing Home Price Index,* unpublished report. Mortgage rates: 1960–62: FHA insured rate plus .44 percentage points. 1963–90: Federal Housing Finance Board, *1990 Rates & Terms on Conventional Home Mortgages, Annual Summary.*

Figure 1.4, this new equilibrium results in a new solution box that is lower and more rectangular than the original.

It is important to be convinced that the new solution box is as portrayed in Figure 1.4. Asset prices must be higher and rents lower, while the long-term stock and its supporting level of construction must be greater. If rents were not lower, the stock would have to be the same (or lower), and this would be inconsistent with higher asset prices and greater construction. If asset prices were not higher, rents would be lower, which is inconsistent with the reduced stock (and less construction) generated by lower asset prices. A positive shift in asset demand, like a positive shift in space demand, will raise prices, construction, and the stock. It will, however, eventually lower—rather than raise—the level of rents. This inverse relationship between asset prices and long-term interest rates can be seen in Figure 1.5, which examines the historic movements in house prices (in 1990 dollars) and real mortgage rates (adjusted for inflation).

Credit, Construction Costs, and the Supply of New Space

The final exogenous change likely to impact the real estate market is a shift in the supply schedule for new construction. This can come about through several channels. Higher short-term interest rates and a general scarcity of construction financing will increase the costs of providing a given amount of new space, leading to less construction. Likewise, stricter local zoning or other building regulations will also add to development costs and (for a fixed level of asset prices) reduce the profitability of new construction. These kinds of negative supply changes have the effect of causing a westerly shift in the cost schedule of the southwest quadrant: for the same level of asset prices, construction will be less. Positive changes in the supply environment, such as the easy availability of construction financing or a relaxation of development regulations, move the curve in an easterly direction, and for the same asset price expand construction. Figure 1.6 shows the powerful inverse relationship that has existed historically between short-term real interest rates (inflation-adjusted) and new home construction.

Finally, Figure 1.7 traces the long-run implications of a negative supply change, such as would occur with higher short-term interest rates.[8] For a given level of asset prices, a negative shift in the new space supply schedule (southwest quadrant) will lower the level of construction and eventually lower the stock of space (southeast quadrant). With less space in the northeast quadrant, rent levels will have to rise, which will generate higher asset prices in the northwest quadrant. When starting and finishing asset prices

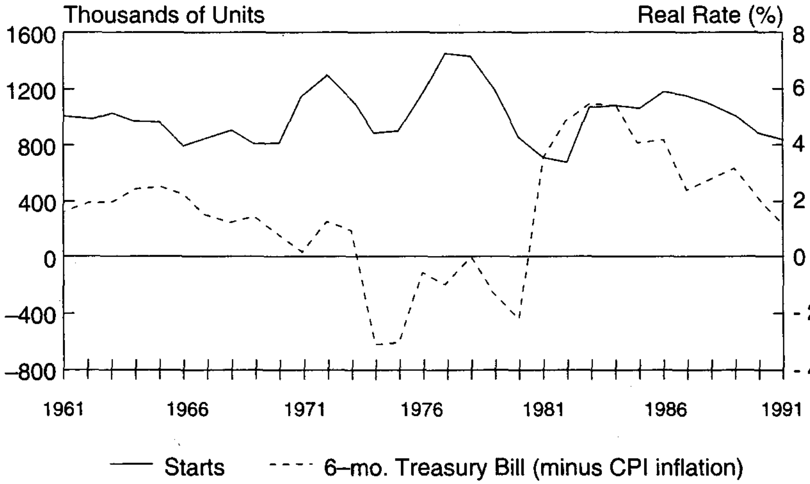

FIGURE 1.6 Single-family construction and real interest rates, 1961–1991.
Source: Construction, *Economic Report of the President 1992*; interest rates, The Federal Reserve.

[8] In Figure 1.7, this negative supply change is drawn as a shift in the cost curve, which means that the minimum level of asset price necessary to get construction going has increased. There may also be supply changes that leave this minimum asset price the same, but lower or raise the amount of construction resulting from increases in asset prices. Such changes would result in a pivot in the schedule, changing its slope (i.e., if the slope becomes flatter there is less construction with increases in asset prices; if the slope becomes steeper, there is more construction with increases in asset prices).

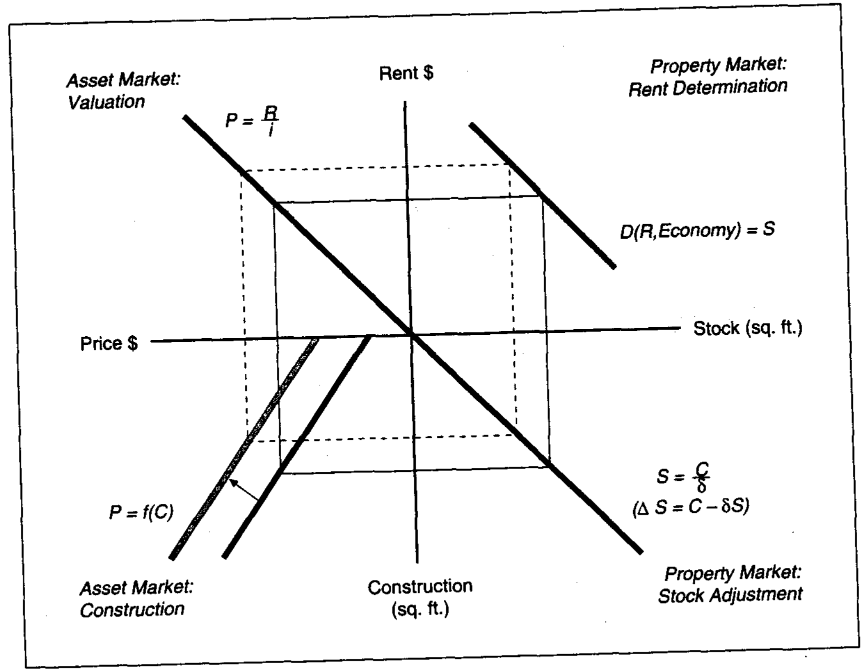

FIGURE 1.7 The property and asset markets: asset cost shifts.

are equal, the new solution box will lie strictly to the northwest of the original solution. Rents and asset prices will increase, whereas construction and stock levels will be less. The magnitude of these changes, of course, will depend on the slopes (or elasticities) of the various curves. For example, if the demand for space is very rent elastic (a nearly horizontal curve in the northeast quadrant), then the increase in rents will be slight. An inelastic (nearly vertical) demand curve will generate a larger increase in rents. It should be clear that if any of these variables moved in a different direction, the solution would be inconsistent—that is, not an equilibrium.

Occasionally, single, individual shifts, such as are portrayed in Figures 1.2, 1.4, and 1.7, do occur by themselves. For example, the Economic Recovery Tax Act of 1981 (ERTA), greatly shortened the depreciation period for rental housing. This generated a sharp reduction in the capitalization rate for this asset, and the ensuing construction boom and fall in rents are quite consistent with Figure 1.4. Some researchers are also convinced that much of the commercial building boom and eventual fall in office rents and asset prices during the 1980s was the result of deregulating the nation's thrift (savings and loan, or S&L) industry. Deregulation is argued to have resulted in a dramatic increase in the availability of cheap construction credit for new commercial development. As such, it would have caused a singular (easterly) shift in the construction cost schedule.

It is more likely the case that economic events cause several shifts to occur simultaneously. This is particularly true of movements in the nation's macroeconomy. As the national economy enters a slowdown, not only is there a contraction in output and employment (northeast quadrant), but there are usually increases in short-term interest rates as well (southwest quadrant). An economic expansion leads to the opposite combination. This combination of shifts can generate any pattern of new box solutions that lies between the two shown in Figures 1.2 and 1.7. Although the analysis gets more complicated in the case of multiple shifts, the net outcome is always some combination of the impacts from each individual change.

The simple framework represented by the four-quadrant diagram works well in illustrating the new equilibria that result as the exogenous environment changes. An important drawback of this framework is that it is not easy to trace the intermediate steps as the market moves to its new equilibrium. A dynamic system of equations is needed to depict the intermediate adjustments of the market, which significantly complicates our analysis. More complicated dynamic models will be developed in Chapters 10 and 12.

REAL ESTATE MARKETS AND PUBLIC POLICY

The real estate sector is affected by public policy changes at the federal, state, and local levels. However, the real estate sector may or may not be the primary target of a specific policy. For example, national monetary policy has several goals and implications. As a prime determinant of interest rate movements, monetary policy clearly affects the demand for real estate assets as well as the level of new construction. The four-quadrant diagrams presented in the previous section illustrate the impacts of such policy changes on the real estate markets. The public sector also creates policies that are aimed specifically at the real estate sector. In developing such policies, government decision makers must be able to anticipate the impacts of their actions on the broader real estate market. The framework represented by the four-quadrant diagram permits a useful analysis of the impacts of these policies. Consider the following examples of major public policies designed to impact the real estate markets directly.

Publicly Assisted Housing

Both federal and state governments provide a range of assistance programs to encourage the development of low- and moderate-income housing. Some programs directly build units for targeted groups, while other programs assist households in making rental payments. The construction of publicly owned units generally decreases the demand for privately owned rental units. This decrease in demand assumes that public units succeed in attracting tenants. This seems a safe assumption since public units generally carry subsidies, which mean that the units are offered at below market rents. In fact, in many urban areas in the U.S., there are long waiting lists for public housing units. This decrease in

demand for private units results in an inward shift in the demand curve in the northeast quadrant which will, in turn, decrease rents, asset prices, construction and ultimately the stock of private units. Thus, building public housing produces a new equilibrium that is exactly the opposite of that illustrated in Figure 1.2, where demand is increasing. The reduction of private building activity due to public construction programs is sometimes referred to as public displacement of private construction.

Rental assistance programs, on the other hand, act to simulate housing demand in much the same way as an expansion in the economy. These programs will shift out the demand curve in the northeast quadrant, resulting in increases in rents, prices, construction, and the stock, as occurred in Figure 1.2. Proponents of rental assistance programs argue that such programs will encourage construction and have only a modest impact on rents. Hence, they argue that the net effect of such programs will be to expand the housing opportunities of low-income households. Opponents of such policies suggest that rental assistance only serves to increase rents in the market and has very little impact on construction. Hence, they argue that the main beneficiaries of the program are landlords. Clearly, the size of the impact on construction and rents depends on the relative elasticities of the curves presented in Figure 1.2.

Local Government Development Regulations

In the U.S., local governments exercise great control over the amount and type of development permitted on privately owned land. Such regulations often are in the public interest, but they do impose two additional costs on private development. First, they frequently extend the time necessary for completion of a project, since local governments require developers to apply for various permits in order to proceed. Second, they sometimes act to create a scarcity of sites. This can drive up land prices and add to site-acquisition costs. The more binding or restrictive such regulations become, the more they increase development costs. This increase in costs will shift the supply schedule in the southwest quadrant in a westerly direction, as occurred in Figure 1.7.

The Taxation of Real Estate

The federal tax code generally treats real estate favorably in several ways. Interest payments on real estate debt are fully tax deductible for both firms and households. Homeowners also are effectively exempt from the taxation of housing capital gains realized with the sale of their homes. For investors, generous depreciation deductions are allowed each year far in excess of actual economic obsolescence. Favorable provisions like these act to reduce the current yield necessary from real estate, as the tax advantage supplements property income. This will rotate the capitalization schedule of the northwest quadrant in a counterclockwise manner and lead to the higher asset prices and other impacts described in Figure 1.4.

At the local level, real estate is treated quite unfavorably, largely through the widespread use of property taxation. Most local governments finance their services with a flat tax rate on commercial, industrial, and residential property value. The effective rate of such taxation generally is in the 1 to 2 percent range, and this directly raises the capitalization rate necessary for real estate. Increased property taxation will rotate the capitalization ray clockwise, reducing asset prices, reducing construction, and raising rents.

Real Estate Financial Institutions and Regulations

In the U.S., the federal government has created a number of institutions whose purpose is to facilitate the financing of residential real estate. The system of S&L banks was established to channel local savings into mortgage lending, and the secondary market together with national mortgage insurance ensure that mortgages have almost instant liquidity. By facilitating the flow of investment funds into mortgage lending, such institutions have effectively reduced the cost of mortgage borrowing. This again will rotate the capitalization rate ray of the northwest quadrant in a counterclockwise manner, increasing asset prices and residential construction.

The federal government also regulates a number of financial institutions in ways that can increase or decrease the flows both of long-term investment funds and short-term construction lending to nonresidential real estate. For example, when the Employee Retirement Income Security Act of 1974 (ERISA) was passed by Congress, pension funds were required to increase diversification of their assets—a policy which many believe greatly increased the supply of funds for long-term commercial mortgages. In 1989, Congress enacted the Financial Institutions Reform, Recovery, and Enforcement Act (FIRREA) which was designed to deal with the S&L crisis. FIRREA increased the capital reserve requirements for loans that were viewed as risky, such as commercial mortgages and short-term construction loans. These requirements substantially decreased the willingness of lenders to originate and hold these loans. The history of the U.S. financial industry is full of many such examples, each of which has influenced the flow of funds into real estate. In our four-quadrant diagram, a decrease in the availability of long-term financing will shift the capitalization rate in the northwest quadrant; a decrease in the availability of short-term construction financing will shift the cost schedule in the southwest quadrant.

SUMMARY

In this chapter, we defined real estate and provided a simple analytic framework to examine the operation of this important market.

- Real estate is defined as the national stock of buildings, the land on which those buildings sit, and all vacant land. Real estate is a very important component of our nation's wealth; the total value of real estate was estimated at $8.8 trillion in 1990, representing 56 percent of the nation's net worth. Additions to the stock represent roughly 5 to 7 percent of annual GDP.
- The four-quadrant model divides the real estate market into the property market where space use is determined and the asset market where buildings are bought and sold. There are two critical links between these markets. First, rents determined in the property market are translated into asset prices in the asset market. Second, asset prices determine the level of new construction, which determines the amount of stock available in the property market.

Exogenous shocks to the property market can have very different impacts on the operation on the real estate market than exogenous shocks to the asset market. In this chapter, we showed that:

- An increase in the demand for space in the property market shifts out the demand curve in the northeast quadrant, increasing rents, which, in turn, increases asset prices, construction, and the stock of space.
- A decrease in the capitalization rate in the asset market increases the demand for real estate assets, which increases asset prices. Increased asset prices in turn bring forth more construction, increasing the stock of space and decreasing rents.
- An increase in construction costs decreases construction levels, which, in turn, decreases the stock of space, driving up both rents and asset prices.

REFERENCES AND ADDITIONAL READINGS

Balance Sheets of the U.S. Economy: 1945–1990. Washington, D.C.: Board of Governors of the Federal Reserve System, September 1991.

BRUEGGEMAN, WILLIAM, J. FISHER, AND LEO STONE, *Real Estate Finance,* 8th ed. Homewood, Ill: Irwin, 1989.

Managing the Future: Real Estate in the 1990s. Chicago: IREM Foundation and Arthur Anderson, 1991.

MILES, MIKE E., "What Is the Value of All U.S. Real Estate?" *Real Estate Review* (1990), 20(2): 69–77.